

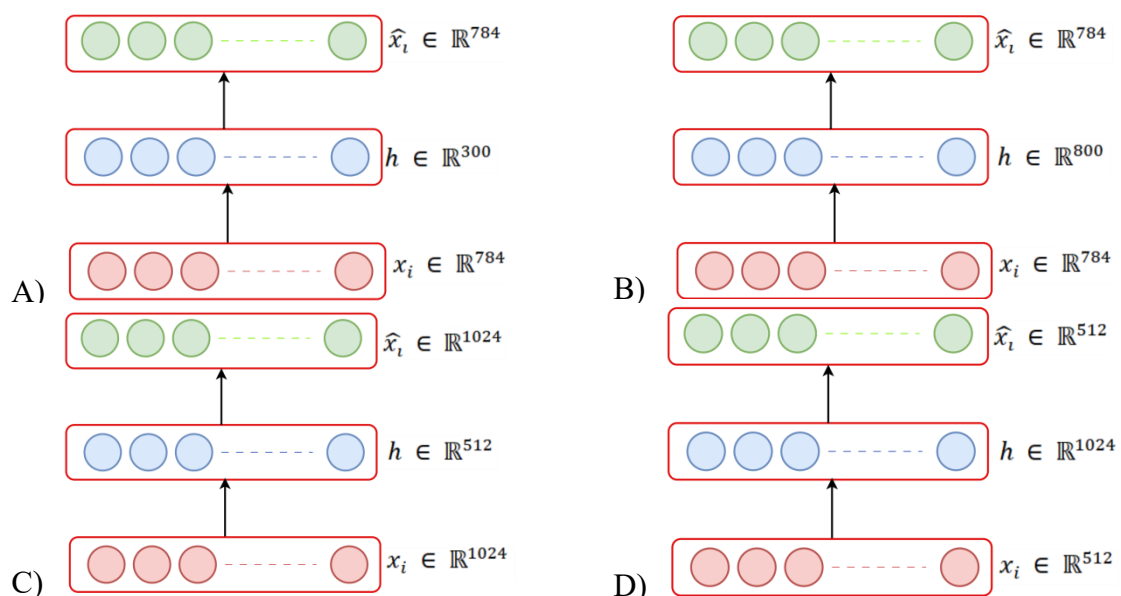
Week 6

Deep Learning

Question: 1

You're working in a healthcare startup building a compression system for chest X-ray images using autoencoders. Your team has proposed the following 4 architectures for evaluation. Each diagram shows the number of neurons in the input layer, hidden layer, and output layer.

Your job is to choose the architecture(s) that will best compress the input while still allowing reasonable reconstruction. Based on the diagrams shown, identify which architecture(s) represent an undercomplete autoencoder.



Options:

- A) A and C
- B) B and D
- C) A, C, and D
- D) Only A

Correct Answer: A

Explanation:

An undercomplete autoencoder has $\dim(h) < \dim(x)$, forcing the network to learn compressed representations. The images A and C have hidden layers smaller than the input.

Data for questions from 2 to 5

A smart factory deploys several sensors across machines to collect temperature, vibration, and humidity readings every second. A deep learning engineer proposes to use an autoencoder to compress and reconstruct this data to detect anomalies based on reconstruction error. The input vector x from sensors is 100-dimensional. The engineer considers two different autoencoder architectures:

Model A: Hidden layer dimension = 20

Model B: Hidden layer dimension = 120

Each model is trained to minimize the reconstruction error between x and its output $\hat{x}$.

Question: 2

Which of the following best describes the role of the hidden representation in this autoencoder setup?

Options:

- A) It stores an exact copy of the input data.
- B) It captures a lower or higher-dimensional encoding useful for reconstruction.
- C) It applies label information to compress data.
- D) It directly computes reconstruction loss.

Correct Answer: B

Explanation: The hidden layer encodes the input into a latent representation h from which the original data is reconstructed.

Question: 3

What can be said about Model A's architecture in this scenario?

Options:

- A) It is an overcomplete autoencoder.
- B) It is an undercomplete autoencoder.
- C) It risks learning an identity function.
- D) It compresses the input data to extract meaningful features.

Correct Answer: B, D

Explanation: Since $\dim(h) < \dim(x_i)$, it's undercomplete and enforces compression, often resulting in meaningful feature extraction.

Question: 4

What is the major risk of using Model B in this situation?

Options:

- A) It might fail to train due to insufficient parameters.
- B) It enforces sparsity and loses data fidelity.
- C) It performs better due to higher capacity.
- D) It could trivially copy input to output without learning meaningful representations.

Correct Answer: D

Explanation: Overcomplete autoencoders can just copy input, which doesn't help in anomaly detection or feature extraction.

Question: 5

Suppose the autoencoder trained on normal sensor data starts showing high reconstruction error on new input data. What does this indicate?

Options:

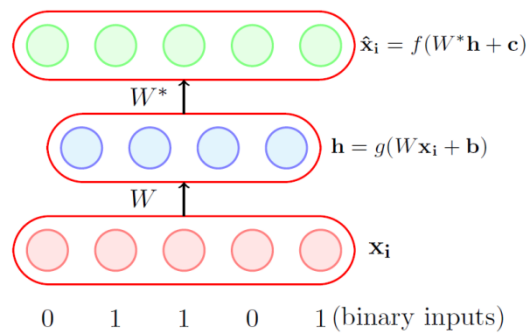
- A) The autoencoder has overfitted.
- B) The sensors are working perfectly.
- C) The new input is likely anomalous or different from training distribution.
- D) The loss function needs to be changed.

Correct Answer: C

Explanation: High reconstruction error suggests the input doesn't align with the learned encoding-decoding patterns—this is often used for anomaly detection.

Data for the questions from 6 to 8

An engineer is designing an autoencoder to compress and reconstruct binary event logs generated by thousands of IoT sensors. Each log vector x is a binary sequence of 64 values, indicating presence (1) or absence (0) of events in each time slot. The goal is to reconstruct the input accurately and use reconstruction error to detect communication failures or faulty sensors.



Question: 6

Which decoder activation function (f) is the most suitable for the above input setting?

Options:

- A) tanh
- B) linear
- C) sigmoid
- D) ReLU

Correct Answer: C

Explanation: The output $\hat{x}$ must lie in $[0,1]$ to match binary inputs. Sigmoid maps outputs into this range, making it suitable for reconstructing binary input.

Question: 7

Which loss function should the engineer use for training the above autoencoder?

Options:

- A) $L(\theta) = \sum_{i=1}^m \|\hat{x}_i - x_i\|_2^2$
- B) $L(\theta) = \sum_{i=1}^m \sum_{j=1}^n -[x_{ij} \log(\hat{x}_{ij}) + (1 - x_{ij}) \log(1 - \hat{x}_{ij})]$
- C) $L(\theta) = \sum_{i=1}^m KL(x_i \| \hat{x}_i)$
- D) $L(\theta) = \sum_{i=1}^m |\hat{x}_i - x_i|$

Correct Answer: B

Explanation: When the input is binary and decoder uses sigmoid, the best match is the binary cross-entropy loss.

Question: 8

Why would using squared error loss with sigmoid decoder and binary input be suboptimal?

Options:

- A) It cannot be used with sigmoid.
- B) It gives poor gradient signals for binary classification tasks.
- C) It assumes Gaussian distribution, which mismatches binary input assumptions.
- D) It leads to unstable training with sigmoid output.

Correct Answer: B, C

Explanation:

Squared error assumes Gaussian output and works poorly with binary data. It also produces less informative gradients when outputs are near 0 or 1.

Data for the questions from 9 to 11

A data science team is building an autoencoder to model daily energy consumption patterns for households in a smart city. Each input vector x is a 24-dimensional real-valued vector, where each element represents electricity usage (in kWh) during one hour of the day.

The objective is to reconstruct the real-valued signal accurately and detect unusual usage patterns via reconstruction error.

Question: 9

Which decoder activation function is most suitable for the given input scenario?

Options:

- A) tanh
- B) linear
- C) sigmoid
- D) ReLU

Correct Answer: B

Explanation: Linear activation allows output in $\mathbb{R}$, matching real-valued inputs. Sigmoid or tanh restricts the range and will clip real values.

Question: 10

Suppose the inputs are normalized but still real-valued. Which loss function best suits the model's objective?

Options:

- A) $L(\theta) = \sum_{i=1}^m \sum_{j=1}^n (\hat{x}_{ij} - x_{ij})^2$
- B) $L(\theta) = \sum_{i=1}^m \sum_{j=1}^n -[x_{ij} \log(\hat{x}_{ij}) + (1 - x_{ij}) \log(1 - \hat{x}_{ij})]$
- C) $L(\theta) = \sum_{i=1}^m KL(x_i || \hat{x}_i)$
- D) $L(\theta) = \sum_{i=1}^m |\hat{x}_i - x_i|$

Correct Answer: A

Explanation: For real-valued inputs, the squared error loss is appropriate as it assumes a Gaussian distribution and penalizes large deviations more.

Question: 11

Why is binary cross-entropy loss generally not appropriate when training an autoencoder on real-valued input data?

Options:

- A) It requires outputs to be strictly positive.
- B) It assumes binary targets in $[0,1]$, which doesn't match real-valued ground truth.
- C) It leads to more accurate reconstructions for continuous data.
- D) It is equivalent to squared loss when using sigmoid outputs.

Correct Answer: B

Explanation: Binary cross-entropy is suitable for binary targets, not continuous real values. Using it with real-valued data violates its probabilistic assumptions and often leads to suboptimal training.

Data for the questions from 12 to 14

A geospatial analytics company handles large volumes of high-resolution satellite imagery data. Each image is represented as a 2048-dimensional real-valued vector after preprocessing. The team wants to reduce dimensionality for downstream clustering and anomaly detection tasks. Two researchers propose different methods:

Researcher A: Use PCA with 100 components.

Researcher B: Train a linear autoencoder with hidden dimension 100, using mean-centered inputs and squared error loss.

Question: 12

Under the above conditions, what can be said about the representations learned by both methods?

Options:

- A) PCA and the linear autoencoder will learn identical subspaces
- B) Autoencoder will learn more nonlinear features
- C) PCA will fail due to lack of labels
- D) Autoencoder needs more computation due to gradient descent

Correct Answers: A, D

Explanation: With linear encoder/decoder, squared error loss, and zero-centered input, the autoencoder approximates PCA. However, PCA solves this via closed-form SVD, while autoencoders require iterative training.

Question: 13

If the dataset is found to lie on a nonlinear manifold, which approach would likely perform better for compression and why?

Options:

- A) PCA, because it captures all the variance
- B) Linear autoencoder, because it approximates PCA
- C) Nonlinear autoencoder, because it can learn nonlinear mappings
- D) None of the above

Correct Answer: C

Explanation: PCA and linear autoencoders both learn linear projections. For nonlinear manifolds, only a nonlinear autoencoder with nonlinear activations can capture the structure better.

Question: 14

The team now wants to visualize the compressed 2D representation of the data. Researcher A uses PCA with 2 components, while Researcher B trains a nonlinear autoencoder with a 2-dimensional hidden layer. Which of the following is true?

Options:

- A) PCA will provide better results since it ensures maximum variance in 2D
- B) The autoencoder may learn better separability due to nonlinearity
- C) Both methods are equivalent for visualization
- D) PCA will outperform because autoencoders need labels

Correct Answers: A, B

Explanation: PCA guarantees maximal variance in linear projections, which is good for visualization. But a nonlinear autoencoder might yield better cluster separation due to its ability to warp the feature space.

Data for questions from 15 to 16

A financial analytics company is using an autoencoder to compress user transaction histories represented as 512-dimensional real-valued vectors. The goal is to detect fraud based on abnormal reconstructions. To increase model capacity, they build a deep autoencoder with hidden layers wider than the input (e.g., $512 \rightarrow 1024 \rightarrow 512 \rightarrow 1024 \rightarrow 512$).

After training, they notice that the model simply copies input to output, resulting in low reconstruction loss even for fraudulent patterns — defeating the purpose.

Question: 15

Why is this trivial copying behavior a problem in overcomplete autoencoders?

Options:

- A) It leads to large training loss
- B) It results in representations that don't generalize
- C) It prevents the decoder from learning the sigmoid function
- D) It increases model interpretability

Correct Answer: B

Explanation: Overcomplete AEs with no constraints tend to learn identity mappings. This prevents extraction of meaningful features and defeats compression.

Question: 16

Which of the following regularization strategies can be used to prevent trivial identity mappings in such overcomplete architectures?

Options:

- A) Adding noise to inputs
- B) Penalizing activations to enforce sparsity
- C) Adding dropout to the output layer only
- D) Penalizing the Jacobian of hidden activations

Correct Answers: A, B, D

Explanation:

Adding noise to inputs

Penalizing activations to enforce sparsity

Penalizing the Jacobian of hidden activations.

Question: 17

A smart city infrastructure team collects air quality readings from thousands of distributed IoT sensors. However, due to packet loss and electromagnetic interference, some sensor readings are partially missing or corrupted. To build a robust model that can learn meaningful patterns despite this corruption, a machine learning engineer trains an autoencoder by randomly corrupting some of the input dimensions and then teaching the model to reconstruct the original uncorrupted input.

What kind of autoencoder is being used in this scenario?

Options:

- A) Sparse Autoencoder

- B) Denoising Autoencoder
- C) Contractive Autoencoder
- D) Overcomplete Autoencoder

Correct Answer: B

Explanation: The model adds noise to the input deliberately and learns to reconstruct the clean version.

Data for questions 18 and 19:

A health-tech startup is building an AI-based image enhancement tool for remote clinics. The clinics upload grayscale X-ray images captured using portable scanners. However, due to hardware limitations and network noise, the received images often contain visual distortions and random noise. To address this, the ML team is exploring autoencoder-based solutions for image denoising.

The input images are of size 128×128 pixels, grayscale. Each image is flattened into a vector before feeding into the model.

Two different autoencoder architectures are proposed:

1. A vanilla autoencoder trained on clean images.
2. A denoising autoencoder (DAE) trained by adding Gaussian noise to the input images and then reconstructing the clean versions.

Question: 18

What is the dimension of the input vector x_i given to the autoencoder?

Fill the blank _____

Correct Answer: 16384 or 16,384

Explanation:

The image is $128 \times 128 = 16,384$ pixels.

Question: 19

Which of the following statements correctly compares the regular and denoising autoencoder approaches for this medical image task?

Options:

- A) Vanilla autoencoder learns to reconstruct clean inputs, but may fail when noisy inputs are given at test time
- B) Denoising autoencoder explicitly learns to remove noise and is better suited for this application
- C) Both autoencoders will perform equally well if trained on clean data

- D) Denoising autoencoders require label supervision, while vanilla autoencoders do not

Correct Answers: A, B

Explanation:

Regular AEs trained on clean data don't generalize well to noisy inputs.
DAEs are trained to handle noisy → clean mapping, ideal for denoising.

Question: 20

An engineer builds an autoencoder for real-valued inputs but uses a sigmoid decoder activation and binary cross-entropy loss. What issue might arise from this setup?

- A) There will be a mismatch between output range and data distribution
- B) The model will have high reconstruction accuracy
- C) Binary cross-entropy is optimal for real-valued targets
- D) There is no issue — this is standard practice

Correct Answer: A

Explanation: Sigmoid activation restricts output to $[0,1]$, and binary cross-entropy is suited for binary targets. This setup is inappropriate for real-valued inputs, where a linear decoder with squared loss is more suitable.

Question: 21

Which of the following loss functions encourages sparsity in hidden unit activations?

Options:

- A) $L(\theta) = \sum_{i=1}^m \|x_i - \hat{x}_i\|^2 + \lambda \sum_{j=1}^n KL(\rho \parallel \hat{\rho}_j)$
- B) $L(\theta) = \sum_{i=1}^m \|x_i - \hat{x}_i\|^2 + \lambda \sum_{j=1}^n \|h_j\|^2$
- C) $L(\theta) = \sum_{i=1}^m KL(x_i \parallel \hat{x}_i)$
- D) $L(\theta) = - \sum x_i \log \hat{x}_i$

Correct Answer: A

Explanation: Option (A) adds a KL divergence term between desired sparsity ρ and the average activation $\hat{\rho}_j$ of each hidden unit, which is the standard regularization used in Sparse Autoencoders.

Question: 22

What is the main idea behind a Sparse Autoencoder?

Options:

- A) It forces the hidden representation to be close to 1

- B) It minimizes the Jacobian of the hidden layer
- C) It restricts most hidden units to be inactive (close to 0)
- D) It enforces that the encoder is always overcomplete

Correct Answer: C

Explanation: Sparse AEs aim to ensure that only a few hidden units are active for any given input by penalizing the average activation using KL divergence or L1 regularization.

Question: 23

Which activation pattern is most likely from a well-trained sparse autoencoder?

Options:

- A) All hidden units have average activation ≈ 0.5
- B) Most hidden units are frequently active across inputs
- C) Hidden units always output 1 or 0
- D) Only a small subset of hidden units fire for each input

Correct Answer: D

Explanation: Sparsity forces selective activation, so only a few neurons should fire per input.

Question: 24

What is the key regularization objective of a Contractive Autoencoder?

Options:

- A) Penalize the sensitivity of the encoder to small input changes
- B) Encourage outputs close to 1
- C) Match the hidden activations to a target probability
- D) Apply noise to the input during training

Correct Answer: A

Explanation: CAEs add a penalty on the Jacobian of the encoder activations w.r.t. input to reduce sensitivity and learn robust representations.

Question: 25

Match the Autoencoder variant with the application it is best suited for:

Autoencoder Variant	Application
A. Vanilla Autoencoder	1. Robust image classification under noisy inputs
B. Denoising Autoencoder	2. Dimensionality reduction for clean sensor data
C. Sparse Autoencoder	3. Feature extraction for gene expression datasets
D. Contractive Autoencoder	4. Learning stable representation invariant to slight input changes.

Options:

- A) A-2, B-1, C-3, D-4
- B) A-3, B-2, C-1, D-4
- C) A-1, B-4, C-2, D-3
- D) A-4, B-1, C-2, D-3

Correct Answer: A

Explanation:

Vanilla AE : Works well on clean data for simple compression tasks.

Denoising AE : Trained to handle noisy inputs and reconstruct clean data — ideal for robustness in noisy environments.

Sparse AE: Sparse representations are useful in high-dimensional, low-sample problems like gene expression data.

Contractive AE: Penalizes input sensitivity → learns invariant features, beneficial for stable recognition tasks.

Question: 26

In a Contractive Autoencoder, the added penalty on the Jacobian of hidden activations with respect to the input encourages which of the following behaviors?

Options:

- A) Encoder output becomes highly sensitive to input changes
- B) Decoder learns to reconstruct noisy images
- C) Input features are transformed into binary values
- D) Hidden representation becomes invariant to small input perturbations

Correct Answer: D

Explanation: CAEs penalize the norm of the encoder's Jacobian $\|\nabla_x h\|^2$, making the learned features robust to small changes in the input, which improves generalization.

Question: 27

Match the following autoencoder setups to their respective types:

Architecture	Autoencoder Type
A. Input: 784 → Hidden: 64 → Output: 784	1. Undercomplete Autoencoder
B. Input: 784 → Hidden: 1024 → Output: 784	2. Overcomplete Autoencoder
C. Input: 784 → Noise added → Hidden: 784 → Output: 784	3. Denoising Autoencoder
D. : 784 → Hidden: 512 with KL divergence on activations → Output: 784	4. Sparse Autoencoder

Options:

(A) A-1, B-2, C-3, D-4

- (B) A-3, B-1, C-4, D-2
- (C) A-4, B-3, C-2, D-1
- (D) A-1, B-4, C-3, D-2

Correct Answer: A

Explanation:

- A: Fewer hidden units → Undercomplete
- B: More hidden units → Overcomplete
- C: Noise added before encoding → Denoising
- D: KL divergence on activations → Sparse AE